

Q 1
2000

Use the substitution $u = 2 + x$ to find $\int \frac{x}{\sqrt{2+x}} dx$.

Let $u = 2 + x$

$$\therefore \frac{du}{dx} = 1$$

$$\therefore du = dx$$

$$\text{and } x = u - 2$$

Now $\int \frac{x}{\sqrt{2+x}} dx$

$$= \int \frac{u-2}{\sqrt{u}} du$$

$$= \int \left(u^{\frac{1}{2}} - 2u^{-\frac{1}{2}} \right) du$$

$$= \frac{2u^{\frac{3}{2}}}{\frac{3}{2}} - 4u^{\frac{1}{2}} + C$$

$$= 2u^{\frac{3}{2}} \left(\frac{u}{3} - 2 \right) + C$$

$$= \frac{2}{3} \sqrt{u} (u - 6) + C$$

$$= \frac{2}{3} \sqrt{2+x} (2+x-6) + C$$

$$= \frac{2}{3} \sqrt{2+x} (x-4) + C.$$

Q 2
2001

Use the substitution $u = 1 + x$ to evaluate

$$15 \int_{-1}^0 x \sqrt{1+x} dx.$$

(f) Let $u = 1 + x$

$$\therefore \frac{du}{dx} = 1$$

$$\therefore du = dx$$

$$\text{and } x = u - 1$$

$$\text{At } x = -1, u = 0$$

$$\text{At } x = 0, u = 1$$

Now $15 \int_{-1}^0 x \sqrt{1+x} dx$

$$= 15 \int_0^1 (u-1) \sqrt{u} du$$

$$= 15 \int_0^1 \left(u^{\frac{3}{2}} - u^{\frac{1}{2}} \right) du$$

$$= 15 \left[\frac{2u^{\frac{5}{2}}}{\frac{5}{2}} - \frac{2u^{\frac{3}{2}}}{\frac{3}{2}} \right]_0^1$$

$$= 15 \left(\frac{2}{5} - \frac{2}{3} - 0 \right)$$

$$= 15 \left(\frac{6-10}{15} \right)$$

$$= -4.$$

3
Q 3
2002

Use the substitution $u = 1 - x^2$ to evaluate $\int_2^3 \frac{2x}{(1-x^2)^2} dx$.

3

(f) Let $u = 1 - x^2$

$$\therefore \frac{du}{dx} = -2x$$

$$\therefore du = -2x dx$$

$$\text{At } x = 2, u = -3$$

$$\text{At } x = 3, u = -8$$

Now $\int_2^3 \frac{2x}{(1-x^2)^2} dx$

$$= \int_{-3}^{-8} \frac{-du}{u^2}$$

$$= - \int_{-3}^{-8} \frac{du}{u^2}$$

$$= - \int_{-3}^{-8} u^{-2} du$$

$$= - \left[\frac{u^{-1}}{-1} \right]_{-3}^{-8}$$

$$= \left[\frac{1}{u} \right]_{-3}^{-8}$$

$$= -\frac{1}{8} - \left(-\frac{1}{3} \right)$$

$$= -\frac{1}{8} + \frac{1}{3}$$

$$= -\frac{3}{24} + \frac{8}{24}$$

$$= \frac{5}{24}.$$





Integration BOS

Part A: Substitution

Q 4

2003

Use the substitution $u = x^2 + 1$ to evaluate

$$\int_0^2 \frac{x}{(x^2 + 1)^3} dx.$$

(e) Let $u = x^2 + 1$

$$\therefore \frac{du}{dx} = 2x$$

$$\therefore du = 2x dx$$

$$\text{At } x = 0, u = 1$$

$$\text{At } x = 2, u = 5$$

$$\text{Now } \int_0^2 \frac{x}{(x^2 + 1)^3} dx$$

$$= \frac{1}{2} \int_0^2 \frac{2x}{(x^2 + 1)^3} dx$$

$$= \frac{1}{2} \int_1^5 \frac{du}{u^3}$$

$$= \frac{1}{2} \int_1^5 u^{-3} du$$

$$= \frac{1}{2} \left[\frac{u^{-2}}{-2} \right]_1^5$$

$$= \frac{1}{2} \left[-\frac{1}{2u^2} \right]_1^5$$

$$= \frac{1}{2} \left[-\frac{1}{50} - \left(-\frac{1}{2}\right) \right]$$

$$= \frac{1}{2} \left(-\frac{1}{50} + \frac{1}{2} \right)$$

$$= \frac{1}{2} \left(-\frac{1}{50} + \frac{25}{50} \right)$$

$$= \frac{1}{2} \times \frac{24}{50}$$

$$= \frac{12}{50}$$

$$= \frac{6}{25}$$

3

Q 5

2004

Use the substitution $u = x - 3$ to evaluate

$$(e) \quad u = x - 3 \quad \therefore x = u + 3 \quad \int_3^4 x\sqrt{x-3} dx.$$

$$\therefore \frac{du}{dx} = 1$$

$$\therefore du = dx$$

$$\text{At } x = 4, u = 1$$

$$\text{At } x = 3, u = 0$$

$$\therefore \int_3^4 x\sqrt{x-3} dx$$

$$= \int_0^1 (u+3)\sqrt{u} du$$

$$= \int_0^1 u^{\frac{3}{2}} + 3u^{\frac{1}{2}} du$$

$$= \left[\frac{2}{5} u^{\frac{5}{2}} + 3 \cdot \frac{2}{3} u^{\frac{3}{2}} \right]_0^1$$

$$= \left[\frac{2}{5} u^{\frac{5}{2}} + 2u^{\frac{3}{2}} \right]_0^1$$

$$= \left(\frac{2}{5} + 2 \right) - (0 + 0)$$

$$= 2\frac{2}{5}.$$

3

Q 6

2005

Using the substitution $u = 2x^2 + 1$, or otherwise, find $\int x(2x^2 + 1)^{\frac{5}{4}} dx$.

3

$$\text{Q1d Let } u = 2x^2 + 1, \quad \frac{du}{dx} = 4x, \text{ or } x = \frac{1}{4} \frac{du}{dx}.$$

$$\therefore \int x(2x^2 + 1)^{\frac{5}{4}} dx = \int \frac{1}{4} u^{\frac{5}{4}} \frac{du}{dx} dx = \int \frac{1}{4} u^{\frac{5}{4}} du$$

$$= \frac{1}{9} u^{\frac{9}{4}} + C = \frac{1}{9} (2x^2 + 1)^{\frac{9}{4}} + C.$$

Q 7
2006

Using the substitution $u = x^4 + 8$, or otherwise, find $\int x^3 \sqrt{x^4 + 8} \, dx$.

Q1b $u = x^4 + 8, \frac{du}{dx} = 4x^3.$

$$\int x^3 \sqrt{x^4 + 8} \, dx = \int \frac{1}{4} \sqrt{u} \frac{du}{dx} \, dx = \int \frac{1}{4} u^{\frac{1}{2}} \, du = \frac{u^{\frac{3}{2}}}{\frac{3}{2}} + c = \frac{(x^4 + 8)^{\frac{3}{2}}}{6} + c$$

Q 8
2007

Use the substitution $u = 25 - x^2$ to evaluate $\int_3^4 \frac{2x}{\sqrt{25 - x^2}} \, dx$.

Q1e Given $u = 25 - x^2, \therefore -\frac{du}{dx} = 2x.$

When $x = 3, u = 16$; when $x = 4, u = 9$.

$$\begin{aligned} \int_3^4 \frac{2x}{\sqrt{25 - x^2}} \, dx &= -\int_3^4 \frac{1}{\sqrt{u}} \frac{du}{dx} \, dx = -\int_{16}^9 u^{-\frac{1}{2}} \, du = \int_9^{16} u^{-\frac{1}{2}} \, du \\ &= \left[2u^{\frac{1}{2}} \right]_9^{16} = 2. \end{aligned}$$

3
Q 9
2008

Use the substitution $u = \log_e x$ to evaluate $\int_e^{e^2} \frac{1}{x(\log_e x)^2} \, dx$.

(a) $u = \ln x, \frac{du}{dx} = \frac{1}{x} \, dx.$

When $x = e, u = 1$; when $x = e^2, u = 2$.

$$\int_1^2 \frac{1}{u^2} \, du = \left[-\frac{1}{u} \right]_1^2 = -\frac{1}{2} - \left(-\frac{1}{1} \right) = \frac{1}{2}.$$

3
Q10
2009

Using the substitution $u = x^3 + 1$, or otherwise, evaluate $\int_0^2 x^2 e^{x^3+1} \, dx$.

(f) Let $u = x^3 + 1, du = 3x^2 \, dx.$

When $x = 0, u = 1$; when $x = 2, u = 9$.

$$\int_0^2 x^2 e^{x^3+1} \, dx = \frac{1}{3} \int_1^9 e^u \, du = \frac{1}{3} [e^u]_1^9 = \frac{e^9 - e}{3}.$$

Q11
2010

Use the substitution $u = 1 - x$ to evaluate $\int_0^1 x\sqrt{1-x} \, dx$.

3

(e) Let $u = 1 - x, du = -dx$.

When $x = 0, u = 1$; when $x = 1, u = 0$.

$$\begin{aligned} \int_0^1 x\sqrt{1-x} \, dx &= \int_1^0 (1-u)\sqrt{u}(-du) \\ &= \int_0^1 (\sqrt{u} - \sqrt{u^3}) \, du \\ &= \left[\frac{2\sqrt{u^3}}{3} - \frac{2\sqrt{u^5}}{5} \right]_0^1 \\ &= \frac{2}{3} - \frac{2}{5} = \frac{4}{15}. \end{aligned}$$

Q12
2011

Using the substitution $u = \sqrt{x}$, evaluate $\int_1^4 \frac{e^{\sqrt{x}}}{\sqrt{x}} \, dx$.

3

(d) $u = \sqrt{x}, du = \frac{1}{2\sqrt{x}} \, dx$.

When $x = 1, u = 1$; when $x = 4, u = 2$.

$$\int_1^4 \frac{e^{\sqrt{x}}}{\sqrt{x}} \, dx = 2 \int_1^2 e^u \, du = 2[e^u]_1^2 = 2(e^2 - e).$$

Q13
2012

Use the substitution $u = 2 - x$ to evaluate $\int_1^2 x(2-x)^5 \, dx$.

3

) $u = 2 - x, du = -dx$.

When $x = 1, u = 1$; when $x = 2, u = 0$.

$$\begin{aligned} I &= \int_1^0 (2-u)u^5(-du) = \int_0^1 (2u^5 - u^6) \, du \\ &= \left[\frac{u^6}{3} - \frac{u^7}{7} \right]_0^1 = \frac{1}{3} - \frac{1}{7} = \frac{4}{21}. \end{aligned}$$

Q14
2013

Which integral is obtained when the substitution $u = 1 + 2x$ is applied to $\int x\sqrt{1+2x} \, dx$?

(A) $\frac{1}{4} \int (u-1)\sqrt{u} \, du$

(B) $\frac{1}{2} \int (u-1)\sqrt{u} \, du$

(C) $\int (u-1)\sqrt{u} \, du$

(D) $2 \int (u-1)\sqrt{u} \, du$

Q15

2014

Evaluate $\int_2^5 \frac{x}{\sqrt{x-1}} dx$ using the substitution $x = u^2 + 1$.

$$x = u^2 + 1 \Rightarrow dx = 2u du$$

So we have our limits

$$x = 5 \Rightarrow u = 2$$

$$x = 2 \Rightarrow u = 1$$

And then substitute

$$I = \int_1^2 \frac{u^2 + 1}{u} \times 2u du$$

$$= 2 \int_1^2 (u + \frac{1}{u}) du$$

$$= 2 \left[\frac{u^2}{2} + \ln u \right]_1^2$$

$$= 2 \left[\frac{4}{2} + \ln 2 - \left(\frac{1}{2} + \ln 1 \right) \right]$$

$$= \frac{20}{3}$$

Q16

2015

Use the substitution $u = 2x - 1$ to evaluate $\int_1^2 \frac{x}{(2x-1)^2} dx$.

$$u = 2x - 1, du = 2dx \text{ and } x = \frac{u+1}{2}$$

$$\int_1^2 \frac{x}{(2x-1)^2} dx = \frac{1}{4} \int_1^3 \frac{(u+1)}{u^2} du$$

$$= \frac{1}{4} \int_1^3 \left(\frac{1}{u} + \frac{1}{u^2} \right) du$$

$$= \frac{1}{4} \left[\ln u - \frac{1}{u} \right]_1^3$$

$$= \frac{1}{4} \left(1 - \frac{1}{3} + \ln 3 \right)$$

$$= \frac{1}{4} \left(\frac{2}{3} + \ln 3 \right)$$

Q17

2016

Use the substitution $u = x - 4$ to find $\int x\sqrt{x-4} dx$.

$$\begin{aligned} \text{Q11b } \int x\sqrt{x-4} dx &= \int (u+4)u^{\frac{1}{2}} du = \frac{2}{5}u^{\frac{5}{2}} + \frac{8}{3}u^{\frac{3}{2}} + c \\ &= \frac{2}{5}(x-4)^{\frac{5}{2}} + \frac{8}{3}(x-4)^{\frac{3}{2}} + c \end{aligned}$$

Q18

2017

Evaluate $\int_0^3 \frac{x}{\sqrt{x+1}} dx$, using the substitution $x = u^2 - 1$.

Question 11 (e)

Sample answer:

$$\int_0^3 \frac{x}{\sqrt{x+1}} dx$$

$$x = u^2 - 1$$

$$\frac{dx}{du} = 2u$$

$$dx = 2u du$$

$$\text{when } x = 0$$

$$u^2 - 1 = 0$$

$$u = 1$$

$$\text{when } x = 3$$

$$u^2 - 1 = 3$$

$$u = 2$$

$$\int_0^3 \frac{x}{\sqrt{x+1}} dx = \int_1^2 \frac{u^2 - 1}{u} \cdot 2u du$$

$$= 2 \int_1^2 (u - \frac{1}{u}) du$$

$$= 2 \left[\frac{u^2}{2} - \ln u \right]_1^2$$

$$= 2 \left[\left(\frac{4}{2} - \ln 2 \right) - \left(\frac{1}{2} - \ln 1 \right) \right]$$

$$= \frac{8}{3}$$

Q19

2018

Evaluate $\int_{-3}^0 \frac{x}{\sqrt{1-x}} dx$, using the substitution $u = 1 - x$.

Sample answer:

$$I = \int_{-3}^0 \frac{x}{\sqrt{1-x}} dx$$

$$= \int_4^1 \frac{1-u}{\sqrt{u}} (-du)$$

$$= \int_1^4 u^{-\frac{1}{2}} - u^{\frac{1}{2}} du$$

$$= \left[2u^{\frac{1}{2}} - \frac{2}{3}u^{\frac{3}{2}} \right]_1^4$$

$$= 2(2) - \frac{2}{3}(8) - \left(2(1) - \frac{2}{3}(1) \right)$$

$$= 4 - 5\frac{1}{3} - 1\frac{1}{3}$$

$$= -2\frac{2}{3}$$

$$\text{Let } u = 1 - x \\ du = -dx$$

$$x = -3 \rightarrow u = 4$$

$$x = 0 \rightarrow u = 1$$

Q20

2019

Use the substitution $u = \cos^2 x$ to evaluate $\int_0^{\frac{\pi}{4}} \frac{\sin 2x}{4 + \cos^2 x} dx$.

$$\int_0^{\frac{\pi}{4}} \frac{\sin 2x}{4 + \cos^2 x} dx$$

$$\text{let } u = \cos^2 x \\ \frac{du}{dx} = -2\cos x \sin x \\ \therefore du = -\sin 2x dx$$

$$\text{when } x = 0 \quad u = 1$$

$$\text{when } x = \frac{\pi}{4} \quad u = \frac{1}{2}$$

$$\therefore \text{The integral} = \int_1^{\frac{1}{2}} \frac{-du}{4+u}$$

$$= \int_{\frac{1}{2}}^1 \frac{du}{4+u}$$

$$= \left[\ln(4+u) \right]_{\frac{1}{2}}^1$$

$$= \ln 5 - \ln \frac{9}{2}$$

$$= \ln \left(\frac{10}{9} \right)$$

3

Q21

2020

(i) Find $\frac{d}{d\theta}(\sin^3 \theta)$.

1

(ii) Use the substitution $x = \tan \theta$ to evaluate $\int_0^1 \frac{x^2}{(1+x^2)^{\frac{5}{2}}} dx$.

4

$$(i) \quad \frac{d}{d\theta}(\sin^3 \theta) = 3\sin^2 \theta \cos \theta$$

(ii) Let $x = \tan \theta$

$$\frac{dx}{d\theta} = \sec^2 \theta$$

$$\text{At } x = 0, \quad \theta = 0$$

$$\text{at } x = 1, \quad \theta = \frac{\pi}{4}$$

$$\therefore \int_0^1 \frac{x^2}{(1+x^2)^{\frac{5}{2}}} dx = \int_0^{\frac{\pi}{4}} \frac{\tan^2 \theta}{(1+\tan^2 \theta)^{\frac{5}{2}}} \sec^2 \theta d\theta \\ = \int_0^{\frac{\pi}{4}} \frac{\tan^2 \theta}{\sec^3 \theta} d\theta$$

$$= \int_0^{\frac{\pi}{4}} \frac{\sin^2 \theta}{\cos^2 \theta} \cos^3 \theta d\theta$$

$$= \int_0^{\frac{\pi}{4}} \sin^2 \theta \cos \theta d\theta$$

$$= \left[\frac{1}{3} \sin^3 \theta \right]_0^{\frac{\pi}{4}} \quad \text{by part (i)}$$

$$= \frac{1}{3} \times \left(\frac{\sqrt{2}}{2} \right)^3$$

$$= \frac{2\sqrt{2}}{3 \times 2^3}$$

$$= \frac{\sqrt{2}}{12}$$

6

6

Q 1

2002

2019

Evaluate $2 \int_0^{\frac{\pi}{4}} \sin^2 4x \, dx$.

$$\begin{aligned} 2 \int_0^{\frac{\pi}{4}} \sin^2 4x \, dx &= 2 \int_0^{\frac{\pi}{4}} \frac{1}{2} (1 - \cos 8x) \, dx \\ &= \int_0^{\frac{\pi}{4}} (1 - \cos 8x) \, dx \\ &= \left[x - \frac{\sin 8x}{8} \right]_0^{\frac{\pi}{4}} \\ &= \left(\frac{\pi}{4} - \frac{\sin 2\pi}{8} \right) - (0 - 0) \\ &= \frac{\pi}{4} - \frac{0}{8} \\ &= \frac{\pi}{4}. \end{aligned}$$

Q 2

2003

2018

Find $\int \cos^2 3x \, dx$.

$$\begin{aligned} \int \cos^2 3x \, dx &= \frac{1}{2} \int (1 + \cos 6x) \, dx \\ &\quad \left(\text{since } \cos^2 x = \frac{1}{2} (1 + \cos 2x) \right) \\ &= \frac{1}{2} \left(x + \frac{1}{6} \sin 6x \right) + C. \end{aligned}$$

Q 3

2004

Find $\int \cos^2 4x \, dx$.

$$\begin{aligned} \int \cos^2 4x \, dx &= \int \frac{1}{2} (1 + \cos 8x) \, dx \\ &= \frac{1}{2} \int (1 + \cos 8x) \, dx \\ &= \frac{1}{2} \left(x + \frac{1}{8} \sin 8x \right) + C. \end{aligned}$$

Q 4

2005

2008

Evaluate $\int_0^{\frac{\pi}{4}} \cos x \sin^2 x \, dx$.

Q4a Let $u = \sin x$, when $x = 0$, $u = 0$; when $x = \frac{\pi}{4}$, $u = \frac{1}{\sqrt{2}}$

$$\frac{du}{dx} = \cos x.$$

$$\int_0^{\frac{\pi}{4}} \cos x \sin^2 x \, dx = \int_0^{\frac{1}{\sqrt{2}}} u^2 \frac{du}{dx} \, dx = \int_0^{\frac{1}{\sqrt{2}}} u^2 \, du = \left[\frac{u^3}{3} \right]_0^{\frac{1}{\sqrt{2}}} = \frac{\sqrt{2}}{12}.$$



Integration BOS

Part B: sin and cos

Q 5

2006

Find $\int_0^{\frac{\pi}{4}} \sin^2 x \, dx$.

$$\begin{aligned} \text{Q3a } \int_0^{\frac{\pi}{4}} \sin^2 x \, dx &= \int_0^{\frac{\pi}{4}} \frac{1}{2} (1 - \cos 2x) \, dx = \left[\frac{1}{2} \left(x - \frac{\sin 2x}{2} \right) \right]_0^{\frac{\pi}{4}} \\ &= \frac{1}{2} \left(\frac{\pi}{4} - \frac{1}{2} \right) = \frac{\pi - 2}{8}. \end{aligned}$$

Q 6

2012

Which expression is equal to $\int \sin^2 3x \, dx$?

(A) $\frac{1}{2} \left(x - \frac{1}{3} \sin 3x \right) + C$

(B) $\frac{1}{2} \left(x + \frac{1}{3} \sin 3x \right) + C$

(C) $\frac{1}{2} \left(x - \frac{1}{6} \sin 6x \right) + C$

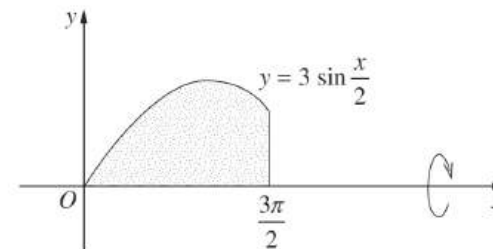
(D) $\frac{1}{2} \left(x + \frac{1}{6} \sin 6x \right) + C$

2

Q 7

2013

The region bounded by the graph $y = 3 \sin \frac{x}{2}$ and the x -axis between $x = 0$ and $x = \frac{3\pi}{2}$ is rotated about the x -axis to form a solid.



Find the exact volume of the solid.

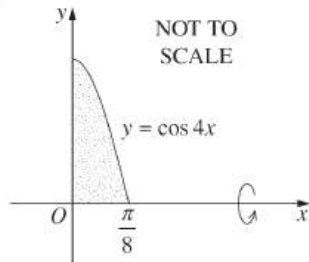
$$\begin{aligned} \text{(b) } V &= \pi \int_0^{\frac{3\pi}{2}} 9 \sin^2 \frac{x}{2} \, dx = \frac{9\pi}{2} \int_0^{\frac{3\pi}{2}} (1 - \cos x) \, dx \\ &= \frac{9\pi}{2} \left[x - \sin x \right]_0^{\frac{3\pi}{2}} = \frac{9\pi}{2} \left(\frac{3\pi}{2} + 1 \right) \text{ u}^3 \end{aligned}$$

Q 8

2014

The region bounded by $y = \cos 4x$ and the x -axis, between $x = 0$ and $x = \frac{\pi}{8}$ is rotated about the x -axis to form a solid.

Find the volume of the solid.



$$\begin{aligned} V &= \pi \int_0^{\frac{\pi}{8}} \cos^2 4x \, dx \\ &= \frac{\pi}{2} \int_0^{\frac{\pi}{8}} (1 + \cos 8x) \, dx \\ &= \frac{\pi}{2} \left[x + \frac{1}{8} \sin 8x \right]_0^{\frac{\pi}{8}} \\ &= \frac{\pi}{2} \left[\frac{\pi}{8} + \frac{1}{8} \sin \pi \right] \\ &= \frac{\pi^2}{16} \end{aligned}$$

Q 9

2015

Find $\int \sin^2 x \, dx$.

$$\int \sin^2 x \, dx = \int \frac{1 - \cos 2x}{2} \, dx = \frac{x}{2} - \frac{\sin 2x}{4} + C$$

Q10

2016

Which expression is equal to $\int \sin^2 2x \, dx$?

- (A) $\frac{1}{2} \left(x - \frac{1}{4} \sin 4x \right) + c$
 (B) $\frac{1}{2} \left(x + \frac{1}{4} \sin 4x \right) + c$
 (C) $\frac{\sin^3 2x}{6} + c$
 (D) $\frac{-\cos^3 2x}{6} + c$

Q11

2020

Find $\int_0^{\frac{\pi}{2}} \cos 5x \sin 3x \, dx$.

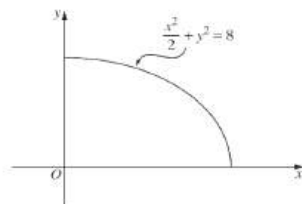
$$\begin{aligned} \int_0^{\frac{\pi}{2}} \cos 5x \sin 3x \, dx &= \int_0^{\frac{\pi}{2}} \frac{1}{2} (\sin 8x - \sin 2x) \, dx \\ &= \left[\frac{1}{2} \left(-\frac{\cos 8x}{8} + \frac{\cos 2x}{2} \right) \right]_0^{\frac{\pi}{2}} \\ &= \frac{1}{2} \left(-\frac{1}{8} - \frac{1}{2} \right) - \frac{1}{2} \left(-\frac{1}{8} + \frac{1}{2} \right) \\ &= -\frac{1}{2} \end{aligned}$$

Q 1

2001

The part of the curve $\frac{x^2}{2} + y^2 = 8$ that lies in the first quadrant is rotated about the x axis.

Find the volume of the solid of revolution.



$$\frac{x^2}{2} + y^2 = 8$$

$$\therefore y^2 = 8 - \frac{x^2}{2}$$

When $y = 0$, $x = 4$.

$$\text{Volume} = \pi \int_a^b y^2 dx$$

$$= \pi \int_0^4 \left(8 - \frac{x^2}{2}\right) dx$$

$$= \pi \left[8x - \frac{x^3}{6} \right]_0^4$$

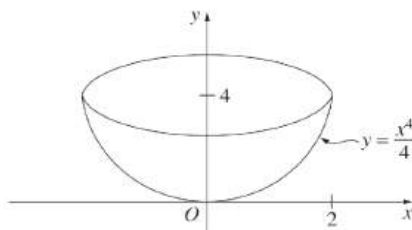
$$= \frac{64\pi}{3} \text{ cubic units.}$$

Q 2

2002

A bowl is formed by rotating the part of the curve $y = \frac{x^4}{4}$ between $x = 0$ and $x = 2$ about the y axis.

Find the volume of the bowl.



$$V = \pi \int_0^4 x^2 dy$$

Now $y = \frac{x^4}{4}$

$$x^4 = 4y$$

$$x^2 = 2\sqrt{y}$$

$$= 2y^{\frac{1}{2}}$$

$$\therefore V = \pi \int_0^4 2y^{\frac{1}{2}} dy$$

$$= 2\pi \int_0^4 y^{\frac{1}{2}} dy$$

$$= 2\pi \left[\frac{2y^{\frac{3}{2}}}{\frac{3}{2}} \right]_0^4$$

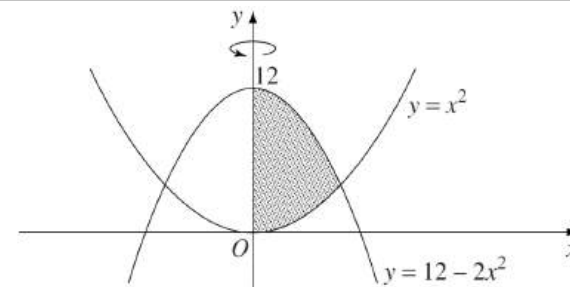
$$= 2\pi \left[\frac{3}{2}(4)^{\frac{3}{2}} - 0 \right]$$

$$= 2\pi \left[\frac{2}{3} \times 8 \right]$$

$$= \frac{32\pi}{3} \text{ units}^3.$$

Q 3

2003



The graphs of the curves $y = x^2$ and $y = 12 - 2x^2$ are shown in the diagram.

- Find the points of intersection of the two curves.
- The shaded region between the curves and the y -axis is rotated about the y -axis. By splitting the shaded region into two parts, or otherwise, find the volume of the solid formed.

(i) Solving simultaneously:

$$y = x^2$$

$$y = 12 - 2x^2$$

then $x^2 = 12 - 2x^2$

$$3x^2 = 12$$

$$x^2 = 4$$

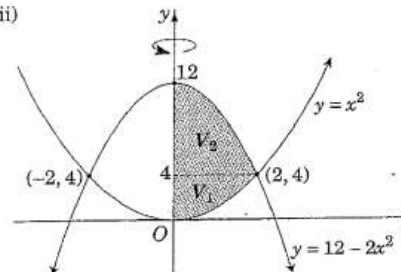
$$x = \pm 2.$$

When $x = -2$, $y = 4$.

When $x = 2$, $y = 4$.

$\therefore$ Points of intersection are $(-2, 4)$ and $(2, 4)$.

(ii)



$$V_{y\text{-axis}} = \pi \int x^2 dy.$$

For $y = 12 - 2x^2$

$$2x^2 = 12 - y$$

$$x^2 = 6 - \frac{y}{2}$$

$$\therefore V = V_1 + V_2$$

$$= \pi \left\{ \int_0^4 y dy + \int_4^{12} \left(6 - \frac{y}{2}\right) dy \right\}$$

$$= \pi \left\{ \left[\frac{y^2}{2} \right]_0^4 + \left[6y - \frac{y^2}{4} \right]_4^{12} \right\}$$

$$= \pi \left\{ \left[\frac{16}{2} - 0 \right] + \left[72 - 36 - (24 - 4) \right] \right\}$$

$$= \pi(8 + 16)$$

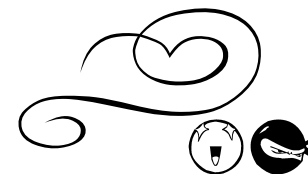
$$= 24\pi \text{ units}^3$$

$$= 75.4 \text{ units}^3 \text{ to 3 sig. figures.}$$



Integration BOS

Part C: Volumes

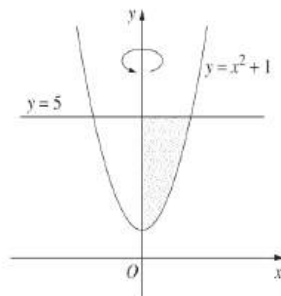


Q 4

2006

In the diagram, the shaded region is bounded by the parabola $y = x^2 + 1$, the y -axis and the line $y = 5$.

Find the volume of the solid formed when the shaded region is rotated about the y -axis.



$$y = x^2 + 1$$

cuts y axis at $x = 0 \therefore (0, 1)$

$$v = \pi \int x^2 dy$$

$$= \pi \int_1^5 y - 1 dy$$

$$= \pi \left[\frac{y^2}{2} - y \right]_1^5$$

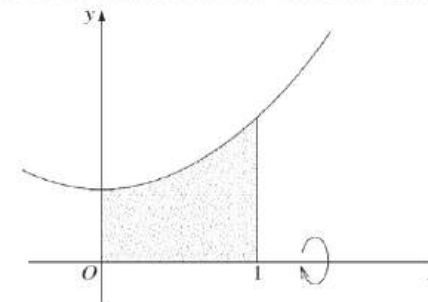
$$= \pi \left[\frac{25}{2} - 5 - \left(\frac{1}{2} - 1 \right) \right] = 8\pi u^3$$

Q 5

2007

The shaded region in the diagram is bounded by the curve $y = x^2 + 1$, the x -axis, and the lines $x = 0$ and $x = 1$.

Find the volume of the solid of revolution formed when the shaded region is rotated about the x -axis.



NOT
TO
SCALE

$$(a) V = \pi \int y^2 dx$$

$$= \pi \int_0^1 (x^2 + 1)^2 dx$$

$$= \pi \int_0^1 x^4 + 2x^2 + 1 dx$$

$$= \pi \left[\frac{x^5}{5} + \frac{2x^3}{3} + x \right]_0^1$$

$$= \pi \left(\frac{1}{5} + \frac{2}{3} + 1 - 0 \right)$$

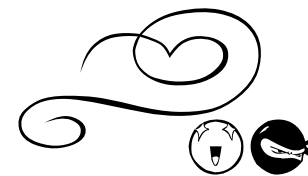
$$= \frac{28}{15} \pi u^3$$





Integration BOS

Part C: Volumes

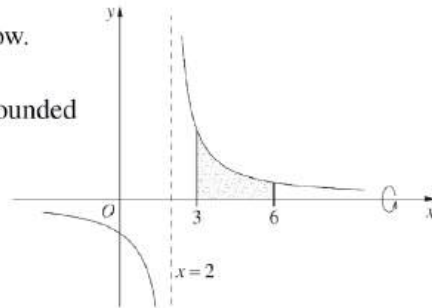


Q 6

2008

The graph of $y = \frac{5}{x-2}$ is shown below.

The shaded region in the diagram is bounded by the curve $y = \frac{5}{x-2}$, the x -axis and the lines $x = 3$ and $x = 6$.



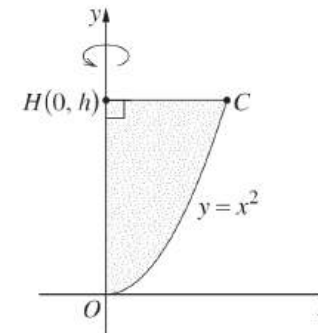
Find the volume of the solid of revolution formed when the shaded region is rotated about the x -axis.

$$\begin{aligned} \text{Q6ci } V &= \int_3^6 \pi y^2 dx = \pi \int_3^6 \frac{25}{(x-2)^2} dx = 25\pi \left[-\frac{1}{x-2} \right]_3^6 \\ &= 25\pi \left(-\frac{1}{4} + 1 \right) = \frac{75\pi}{4} \text{ cubic units.} \end{aligned}$$

Q 7

2011

The diagram shows the region enclosed by the parabola $y = x^2$, the y -axis and the line $y = h$, where $h > 0$. This region is rotated about the y -axis to form a solid called a paraboloid. The point C is the intersection of $y = x^2$ and $y = h$. The point H has coordinates $(0, h)$.



- Find the exact volume of the paraboloid in terms of h .
- A cylinder has radius HC and height h .

What is the ratio of the volume of the paraboloid to the volume of the cylinder?

$$\text{Q8bi } V = \int_0^h \pi x^2 dy = \int_0^h \pi y dy = \left[\frac{\pi y^2}{2} \right]_0^h = \frac{1}{2} \pi h^2$$

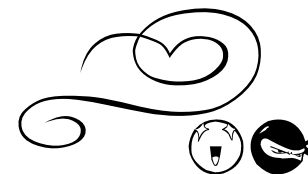
$$\text{Q8bii } V_{\text{cylinder}} = \pi (\sqrt{h})^2 h = \pi h^2$$

$$\therefore \frac{V_{\text{paraboloid}}}{V_{\text{cylinder}}} = \frac{\frac{1}{2} \pi h^2}{\pi h^2} = \frac{1}{2}$$



Integration BOS

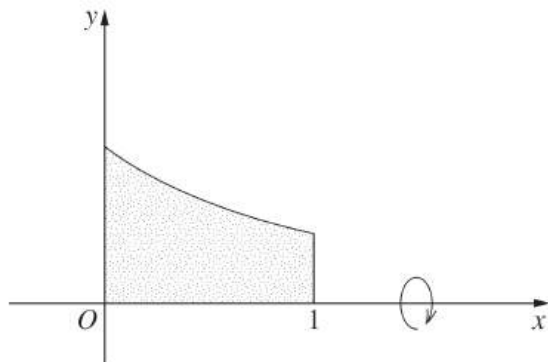
Part C: Volumes



Q 8

2012

The diagram shows the region bounded by $y = \frac{3}{(x+2)^2}$, the x -axis, the y -axis, and the line $x = 1$.



The region is rotated about the x -axis to form a solid.

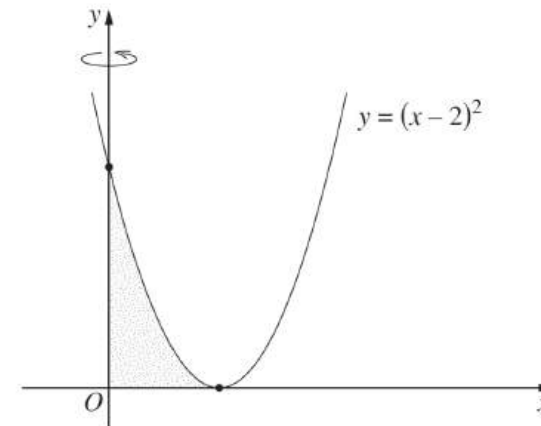
Find the volume of the solid.

$$\begin{aligned} \text{Q14b Volume} &= \int_0^1 \pi \left(\frac{3}{(x+2)^2} \right)^2 dx = \int_0^1 \pi \frac{9}{(x+2)^4} dx \\ &= 9\pi \int_0^1 (x+2)^{-4} dx = 9\pi \left[-\frac{1}{3(x+2)^3} \right]_0^1 = 9\pi \left(-\frac{1}{81} + \frac{1}{24} \right) \\ &= \frac{19\pi}{72}. \end{aligned}$$

Q 9

2013

The region bounded by the x -axis, the y -axis and the parabola $y = (x-2)^2$ is rotated about the y -axis to form a solid.



Find the volume of the solid.

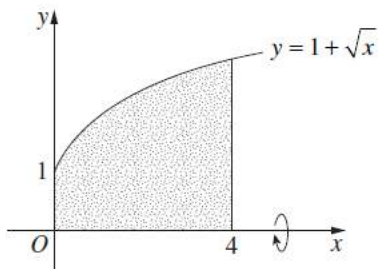
$$\begin{aligned} \text{b) } y &= (x-2)^2 \\ \pm\sqrt{y} &= x-2 \\ x &= 2 \pm \sqrt{y} \\ x &= 2 - \sqrt{y} \text{ valid branch} \\ (\text{sub. } y=1, x=1) \end{aligned}$$

$$\begin{aligned} V &= \pi \int_0^4 (2 - \sqrt{y})^2 dy = \pi \int_0^4 4 - 4\sqrt{y} + y dy = \\ &= \pi \left[4y - \frac{4y^{3/2}}{3/2} + \frac{y^2}{2} \right]_0^4 = \pi \left(16 - \frac{64}{3} + \frac{16}{2} \right) \\ &= \frac{8\pi}{3} \text{ units}^3 \end{aligned}$$

Q10

2014

The region bounded by the curve $y = 1 + \sqrt{x}$ and the x -axis between $x = 0$ and $x = 4$ is rotated about the x -axis to form a solid.



Find the volume of the solid.

$$\begin{aligned} \text{Q14c Volume} &= \int_0^4 \pi y^2 dx = \pi \int_0^4 (1 + 2\sqrt{x} + x) dx \\ &= \pi \left[x + \frac{4x^{\frac{3}{2}}}{3} + \frac{x^2}{2} \right]_0^4 = \frac{68\pi}{3} \text{ cubic units} \end{aligned}$$

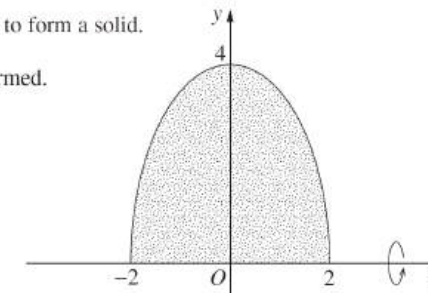
Q11

2017

The diagram shows the region bounded by $y = \sqrt{16 - 4x^2}$ and the x -axis.

The region is rotated about the x -axis to form a solid.

Find the exact volume of the solid formed.



$$\begin{aligned} V &= \pi \int_{-2}^2 (\sqrt{16 - 4x^2})^2 dx \\ &= 2\pi \int_0^2 (16 - 4x^2) dx \\ &= 2\pi \left[16x - \frac{4x^3}{3} \right]_0^2 \\ &= 2\pi \left[32 - \frac{32}{3} - 0 + 0 \right] \end{aligned}$$

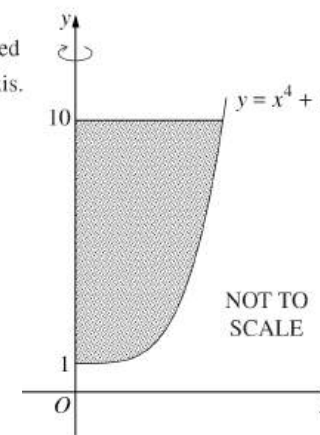
$$\text{Volume} = \frac{128\pi}{3} \text{ units}^3$$

Q12

2018

The shaded region shown in the diagram is bounded by the curve $y = x^4 + 1$, the y -axis and the line $y = 10$.

Find the volume of the solid of revolution formed when the shaded region is rotated about the y -axis.



Sample answer:

$$\begin{aligned} y &= x^4 + 1 \\ x^4 &= y - 1 \\ x^2 &= \sqrt{y - 1} \end{aligned}$$

$$\begin{aligned} V &= \pi \int_1^{10} x^2 dy \\ &= \pi \int_1^{10} (y - 1)^{\frac{1}{2}} dy \end{aligned}$$

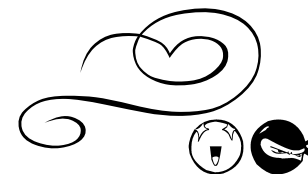
$$\begin{aligned} &= \pi \left[\frac{2(y - 1)^{\frac{3}{2}}}{\frac{3}{2} \times 1} \right]_1^{10} \\ &= \frac{2\pi}{3} \left(9^{\frac{3}{2}} - 0 \right) \\ &= \frac{2\pi}{3} \times 27 \end{aligned}$$

$$\text{Volume} = 18\pi \text{ cubic units}$$



Integration BOS

Part C: Volumes

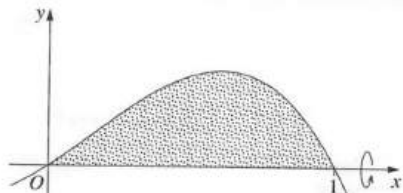


Q13

2019

The diagram shows the region bounded by the curve $y = x - x^3$, and the x -axis between $x = 0$ and $x = 1$. The region is rotated about the x -axis to form a solid.

Find the exact value of the volume of the solid formed.

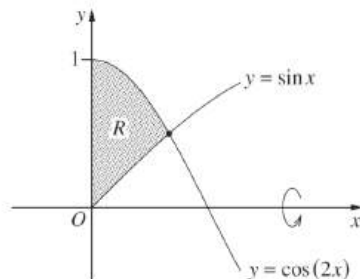


$$\begin{aligned} \text{Q13d Volume} &= \int_0^1 \pi(x - x^3)^2 dx = \pi \int_0^1 x^2 - 2x^4 + x^6 dx \\ &= \pi \left[\frac{x^3}{3} - \frac{2x^5}{5} + \frac{x^7}{7} \right]_0^1 = \frac{8\pi}{105} \end{aligned}$$

Q14

2020

The region R is bounded by the y -axis, the graph of $y = \cos(2x)$ and the graph of $y = \sin x$, as shown in the diagram.



Find the volume of the solid of revolution formed when the region R is rotated about the x -axis.

The curves intersect when

$$\cos 2x = \sin x$$

$$1 - 2\sin^2 x = \sin x$$

$$2\sin^2 x + \sin x - 1 = 0$$

$$(2\sin x - 1)(\sin x + 1) = 0$$

$$\therefore \sin x = \frac{1}{2}, (\sin x \neq -1)$$

$$\therefore x = \frac{\pi}{6}$$

The volume is given by

$$V = \pi \int_0^{\frac{\pi}{6}} \cos^2 2x dx - \pi \int_0^{\frac{\pi}{6}} \sin^2 x dx$$

$$= \pi \left(\int_0^{\frac{\pi}{6}} \frac{1 + \cos 4x}{2} dx - \int_0^{\frac{\pi}{6}} \frac{1 - \cos 2x}{2} dx \right)$$

$$= \pi \left[\frac{\sin 4x}{8} + \frac{x}{2} - \frac{x}{2} + \frac{\sin 2x}{4} \right]_0^{\frac{\pi}{6}}$$

$$= \pi \left(\frac{\sin \frac{2\pi}{3}}{8} + \frac{\sin \frac{\pi}{3}}{4} \right)$$

$$= \frac{\pi}{8} \left(\frac{\sqrt{3}}{2} + \sqrt{3} \right)$$

$$= \frac{3\pi\sqrt{3}}{16}$$

